

2-D Unstructured Compressible Navier–Stokes Solver Benchmark

Reproducible solver submission

August 28, 2026

Abstract

This report documents a C++17/MPI cell-centered finite-volume solver for the eight supplied NACA0012 and circular-cylinder cases. CGNS meshes are partitioned by METIS; rank-local owned/ghost data are advanced with neighbor halo exchange, Rusanov fluxes, least-squares reconstruction, positivity control, implicit local block-Jacobi updates, and a BDF2 dual-time transient loop. All eight required case packages are present and pass the local evidence audit. The recorded status set contains seven converged steady packages and one statistically periodic transient package. Force coefficients, residual reductions, post-transient statistics, partition counts, and every figure are generated from the submitted output files.

1 Introduction

The benchmark evaluates an original two-dimensional compressible Navier–Stokes finite-volume implementation, including unstructured CGNS mesh handling, METIS domain decomposition, limited reconstruction, implicit steady marching, and a true second-order dual-time transient solve. The required set comprises three inviscid NACA0012 cases at $M_\infty = 0.15, 0.8, 2.0$, three laminar NACA0012 cases at $Re = 5000$, a laminar cylinder at $Re = 20$, and the $Re = 200$ vortex-street case. Every status claim below is tied to a result package and to the machine-readable audits.

2 Governing Equations and Nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and the equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}. \quad (1)$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \quad (2)$$

The perfect-gas closure is $p = (\gamma - 1)\rho e$, $E = e + \frac{1}{2}(u^2 + v^2)$, and $a = \sqrt{\gamma p / \rho}$. Reference quantities are ρ_∞ , U_∞ , L_{ref} , p_∞ , $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$, and A_{ref} ; $C_D = D/(q_\infty A_{ref})$, $C_L = L/(q_\infty A_{ref})$, and $C_m = M_z/(q_\infty A_{ref} L_{ref})$. For laminar cases, $\mu = \rho_\infty U_\infty L_{ref} / Re$, $\mathbf{q} = -k\nabla T$, and $k = \mu c_p / Pr$. The Newtonian stresses are

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x). \quad (3)$$

3 Meshes, Boundary Tags, and Case Inputs

The CGNS reader merges coincident zone points, constructs polygonal cell geometry and oriented faces, pairs cylinder interface sections as interior faces, and preserves physical boundary-family names. JSON boundary mappings select farfield, slip-wall, and no-slip adiabatic treatments; no case identifier is hard-coded in the C++ solver. Table 1 records the production mesh sizes and input parameters.

Table 1: Mesh sizes and supplied case parameters.

Case	mesh	N_c	N_f	M_∞	Re	mode	boundary families
naca0012_m015_inviscid	NACA0012_H2.cgn	20816	36498	0.15	0	inviscid	bc-2: farfield; bc-4: slip_wall
naca0012_m080_inviscid	NACA0012_H2.cgn	20816	36498	0.8	0	inviscid	bc-2: farfield; bc-4: slip_wall
naca0012_m200_inviscid	NACA0012_H2.cgn	20816	36498	2	0	inviscid	bc-2: farfield; bc-4: slip_wall
naca0012_m015_laminar_re5000	NACA0012_H2.cgn	20816	36498	0.15	5000	laminar	bc-2: farfield; bc-4: no_slip_adiabatic_wall
naca0012_m080_laminar_re5000	NACA0012_H2.cgn	20816	36498	0.8	5000	laminar	bc-2: farfield; bc-4: no_slip_adiabatic_wall
naca0012_m200_laminar_re5000	NACA0012_H2.cgn	20816	36498	2	5000	laminar	bc-2: farfield; bc-4: no_slip_adiabatic_wall
cylinder_m010_laminar_re20	CylinderB1.cgns	10185	20180	0.1	20	laminar	WALL: no_slip_adiabatic_wall; FAR: farfield
cylinder_m010_laminar_re200	CylinderB1.cgns	10185	20180	0.1	200	laminar	WALL: no_slip_adiabatic_wall; FAR: farfield

4 Spatial Discretization

For owned cell i , the cell-centered finite-volume residual sums oriented face contributions, $\mathcal{R}_i = \sum_{f \in \partial i} |f| [\hat{\mathbf{F}}^i - \hat{\mathbf{F}}^v]_f$, with the face normal directed out of the left cell. Interior faces contribute with opposite signs to their adjacent cells; a rank evaluates a face whenever at least one adjacent cell is owned.

4.1 Second-Order Reconstruction

Primitive variables use an unweighted least-squares gradient and piecewise-linear face states, $\mathbf{q}_f = \mathbf{q}_i + \alpha_i (\nabla \mathbf{q})_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$. In steady production runs the high-order sweep is active on the first inner sweep of every 25th outer step; the remaining steady inner sweeps use a first-order predictor for robustness. In the Re 200 transient run the high-order path is active at every inner residual evaluation. This exact fallback schedule is stated so the reported second-order capability is not confused with an all-sweeps claim.

4.2 Limiter and Positivity Control

The steady high-order path uses a Barth–Jespersen coefficient from neighbor minima and maxima of density and pressure. Reconstructed density and pressure are floored positive. Each conservative update then uses a convex backtracking line search that accepts only finite, admissible density, pressure, speed, and energy; if all 18 halvings fail, the update is rejected. The transient coefficient is unity, but the same admissible-state fallback remains active.

4.3 Inviscid and Viscous Fluxes

The interface flux is local Lax–Friedrichs/Rusanov, $\hat{\mathbf{F}} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2} \max(|u_n| + a)(\mathbf{U}_R - \mathbf{U}_L)$, with the case dissipation scale. Viscous stresses and heat flux use reconstructed primitive gradients. At a no-slip wall, the one-sided normal velocity derivative is imposed from wall distance, tangential gradients are retained, and pressure and tangential skin-friction forces are accumulated in separate columns.

5 Boundary Conditions

Farfield faces use the freestream state through the Rusanov flux. Inviscid slip walls mirror only the normal velocity, giving zero normal mass flux while retaining tangential motion. Viscous walls use a mirrored

inviscid normal state plus Newtonian/Fourier wall flux, zero wall velocity, and zero normal temperature gradient. Surface output reports boundary values: no-slip rows have zero velocity and Mach, whereas slip rows remove only the normal component. The wall gates in `sanity_checks.json` verify these semantics.

6 Implicit and Transient Time Integration

Steady cases use local pseudo-time steps based on convective spectral radii and an additional viscous perimeter/area contribution. Each outer step executes the configured local block-Jacobi relaxation budget while CFL ramps from the supplied initial value to its supplied cap. Convergence requires the requested global residual reduction; a diagnostic `--max-steps` run cannot claim a completed plateau. The Re 200 case uses BDF2 dual time integration, $[1.5 \mathbf{U}^{n+1} - 2 \mathbf{U}^n + 0.5 \mathbf{U}^{n-1}]/\Delta t + \mathcal{R}(\mathbf{U}^{n+1}) = 0$, after backward-Euler startup. The physical-time loop is outermost and the nonlinear pseudo-time loop is inner. $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ remain frozen throughout each inner solve and history shifts occur only after the step meets the total-residual target. Tables 2 and 3 give supplied and observed controls.

Table 2: Supplied production run controls read from the case JSON.

Case	type	max steps	Δt	t_f	CFL_0	CFL_{max}	ramp	inner min-max / target
naca0012_m015_inviscid	steady	20000	–	–	1	100	2000	3–50 / 0.01
naca0012_m080_inviscid	steady	30000	–	–	1	100	3000	3–50 / 0.01
naca0012_m200_inviscid	steady	40000	–	–	0.2	50	5000	3–80 / 0.01
naca0012_m015_laminar_re5000	steady	40000	–	–	1	100	5000	3–80 / 0.01
naca0012_m080_laminar_re5000	steady	40000	–	–	0.5	80	5000	3–80 / 0.01
naca0012_m200_laminar_re5000	steady	50000	–	–	0.2	50	7000	3–100 / 0.01
cylinder_m010_laminar_re20	steady	30000	–	–	1	100	3000	3–80 / 0.01
cylinder_m010_laminar_re200	transient	–	0.01	300	1	1	0	5–1000 / 0.001

Table 3: Observed inner-solve statistics for the transient production case.

Case	observed min	observed mean	observed max	target misses	converged fraction
cylinder_m010_laminar_re200	8	14.973	193	0	1

7 MPI Parallelization and METIS Partitioning

Rank zero reads the CGNS mesh once, builds the cell adjacency graph, and calls METIS K-way partitioning. It writes compact per-rank files containing owned cells, one-ring ghost cells, local faces, and deduplicated send/receive lists. During iterations each rank stores only owned-plus-ghost geometry and states; nonblocking point-to-point exchanges are restricted to neighboring ranks before reconstruction. Residual, force, and statistics use MPI reductions. The global mesh/state is gathered only after the final step for rank-zero field, surface, and restart output, not during iteration. Table 4 reports actual edge cuts and load balance.

Table 4: Per-run METIS partition diagnostics.

Case	owned min-max	ghost total	neighbors mean	edge cut	send/recv	load ratio
naca0012_m015_inviscid	2568–2648	844	3.25	428	856/856	1.018
naca0012_m080_inviscid	2568–2648	844	3.25	428	856/856	1.018
naca0012_m200_inviscid	2568–2648	844	3.25	428	856/856	1.018
naca0012_m015_laminar_re5000	2568–2648	844	3.25	428	856/856	1.018
naca0012_m080_laminar_re5000	2568–2648	844	3.25	428	856/856	1.018
naca0012_m200_laminar_re5000	2568–2648	844	3.25	428	856/856	1.018
cylinder_m010_laminar_re20	1246–1291	961	4.50	497	994/994	1.014
cylinder_m010_laminar_re200	1246–1291	961	4.50	497	994/994	1.014

8 Output, Reproducibility, and Sanity Checks

The release build and production command are documented in `solver/README.md`; a representative invocation is:

```
cmake -S solver -B build -DCMAKE_BUILD_TYPE=Release -DCFD_EXTERNALS_ROOT=external/cfd_externals/i
cmake --build build -j
mpirun -np 8 build/cfd_solver solve --case cfd_solver_agentic_benchmark/inputs/cases/<case>.json
```

Each package contains metadata, per-rank partition diagnostics, residual and force histories, a boundary surface file, an ASCII unstructured field, a restart, standard output, and a run status. Exact commands, ranks, timings, and statuses are in `run_manifest.csv`; every figure is linked to a source CSV or field file in `figure_manifest.csv`; positivity, wall, force-variation, periodicity, MPI-rank, and figure-variable gates are in `sanity_checks.json`.

9 Results

Tables 5 and 6 summarize the production histories. Each case subsection introduces and cross-references panels generated directly from the submitted residual, force, surface, and field files.

9.1 Summary Tables

Table 5: Production run status and residual evidence.

Case	n_p	steps	t_f	orders	wall [s]	status
naca0012_m015_inviscid	8	823	0	4.03211	49.3494	converged
naca0012_m080_inviscid	8	543	0	4.01517	29.898	converged
naca0012_m200_inviscid	8	238	0	3.00953	22.4308	converged
naca0012_m015_laminar_re5000	8	968	0	4.00553	85.8684	converged
naca0012_m080_laminar_re5000	8	5043	0	4.00026	438.995	converged
naca0012_m200_laminar_re5000	8	2529	0	3.00002	277.966	converged
cylinder_m010_laminar_re20	8	542	0	5.15908	30.7102	converged
cylinder_m010_laminar_re200	8	30000	300	1.01932	542.18	statistically-periodic

Table 6: Tail-mean (steady) or post-transient mean (Re 200) force coefficients.

Case	C_D	C_L	C_m	$C_{D,p}$	$C_{D,v}$	$C_{L,p}$	$C_{L,v}$	sta
naca0012_m015_inviscid	0.06588142	-0.0001358899	-6.36691e-05	0.06588142	0	-0.0001358899	0	
naca0012_m080_inviscid	0.0509945	-0.001620908	-0.0006415307	0.0509945	0	-0.001620908	0	
naca0012_m200_inviscid	0.09377743	-0.001038561	-0.00047275	0.09377743	0	-0.001038561	0	
naca0012_m015_laminar_re5000	3.18719	-0.001213959	-0.0007241622	0.1005784	3.086611	-0.001195434	-1.852493e-05	
naca0012_m080_laminar_re5000	0.6924408	-0.002168069	-0.0009840017	0.06126372	0.6311771	-0.002147428	-2.064139e-05	
naca0012_m200_laminar_re5000	0.3286775	-0.001271425	-0.0005967451	0.0950783	0.2335991	-0.00124287	-2.855478e-05	
cylinder_m010_laminar_re20	4.701114	-0.001426419	1.125122e-15	3.4945	1.206615	-0.001080334	-0.0003460853	
cylinder_m010_laminar_re200	1.274232	0.01901239	1.156502e-15	0.8723635	0.401869	0.01610419	0.002908192	

9.2 NACA0012 Cases

The zero-angle NACA cases are assessed for symmetry, nontrivial pressure loading, compressibility, and (for laminar cases) wall shear. The Mach-2 case is expected to show shock/expansion structure; the surface and field plots below use near-body windows.

9.3 naca0012_m015_inviscid

This inviscid slip-wall result is recorded as **converged**; the evidence audit is **converged**. The tail/final values are $C_D = 0.06588142$, $C_L = -0.0001358899$, and $C_m = -6.36691e - 05$ at $M_\infty = 0.15$ and $\alpha = 0^\circ$. The residual reduction is 4.03211 orders; the near-zero lift and wall-condition gates provide the symmetry and boundary checks. The production method is `least_squares_piecewise_linear` with `Barth_Jespersen_positivity_fallback`, `Rusanov_local_Lax_Friedrichs` flux, and `implicit_pseudo_time/local_block_Jacobi_v` integration. The panels fig. 1a, fig. 1b show the submitted source histories or computed fields named in their captions. The panels fig. 2a, fig. 2b show the submitted source histories or computed fields named

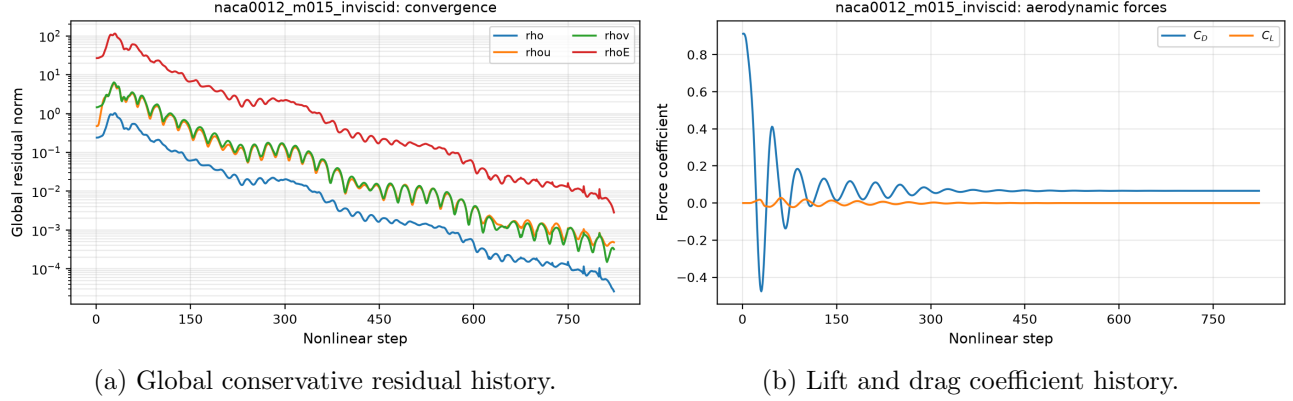


Figure 1: naca0012_m015_inviscid computed-output diagnostics

in their captions. The panels fig. 3a show the submitted source histories or computed fields named in

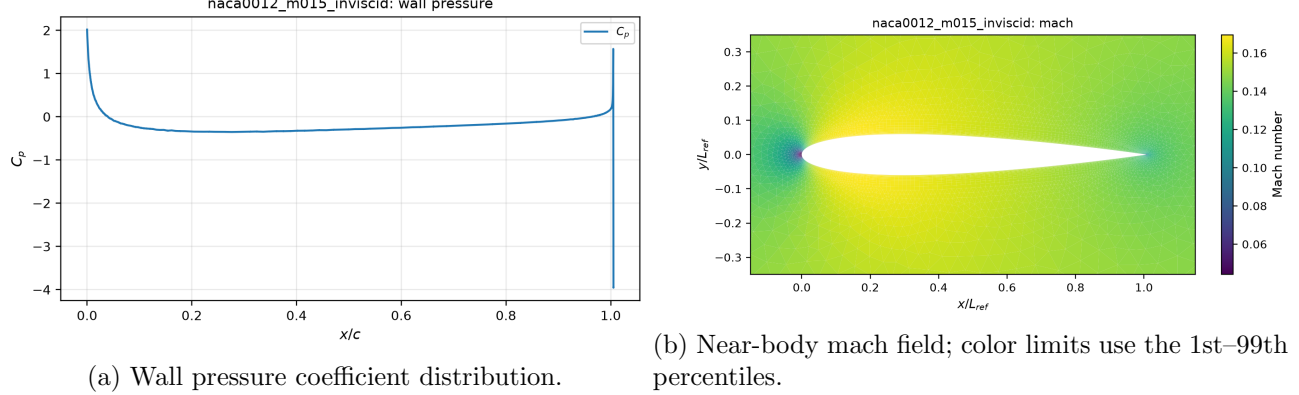
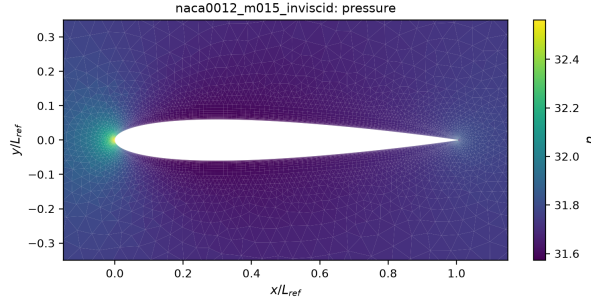


Figure 2: naca0012_m015_inviscid computed-output diagnostics

their captions.

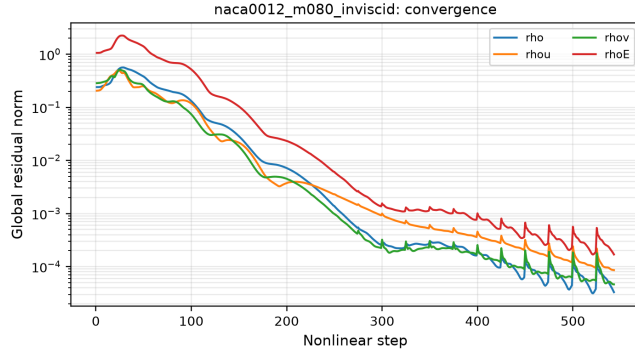
9.4 naca0012_m080_inviscid

This inviscid slip-wall result is recorded as **converged**; the evidence audit is **converged**. The tail/final values are $C_D = 0.0509945$, $C_L = -0.001620908$, and $C_m = -0.0006415307$ at $M_\infty = 0.8$ and $\alpha = 0^\circ$. The residual reduction is 4.01517 orders; the near-zero lift and wall-condition gates provide the symmetry and boundary checks. The production method is `least_squares_piecewise_linear` with `Barth_Jespersen_positivity_fallback`, `Rusanov_local_Lax_Friedrichs` flux, and `implicit_pseudo_time/local_block_Jacobi_with_CFL` integration. The panels fig. 4a, fig. 4b show the submitted source histories or computed fields named in their captions. The panels fig. 5a, fig. 5b show the submitted source histories or computed fields named in their captions. The panels fig. 6a show the submitted source histories or computed fields named in their captions.

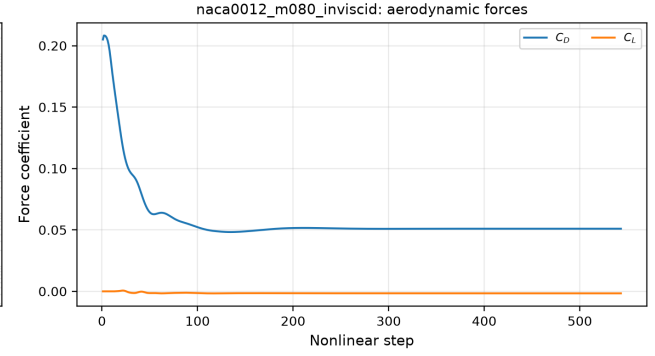


(a) Near-body pressure field; color limits use the 1st–99th percentiles.

Figure 3: naca0012_m015_inviscid computed-output diagnostics

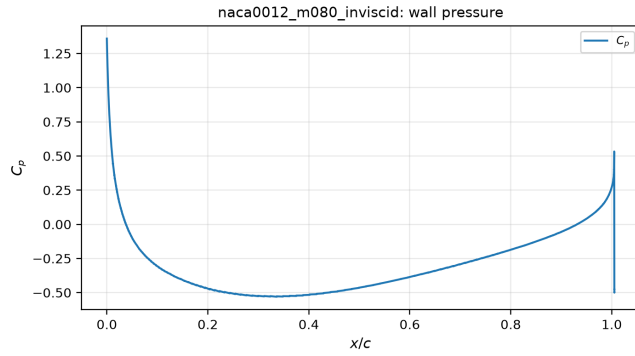


(a) Global conservative residual history.

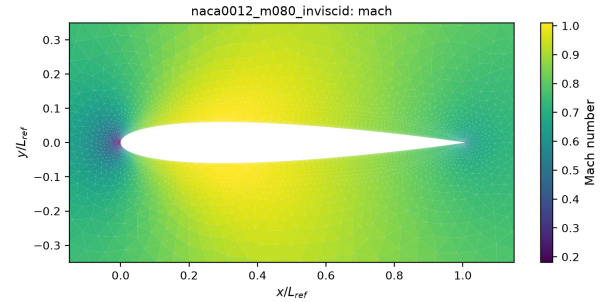


(b) Lift and drag coefficient history.

Figure 4: naca0012_m080_inviscid computed-output diagnostics

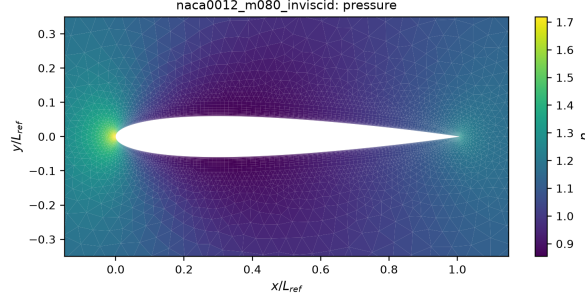


(a) Wall pressure coefficient distribution.



(b) Near-body mach field; color limits use the 1st–99th percentiles.

Figure 5: naca0012_m080_inviscid computed-output diagnostics

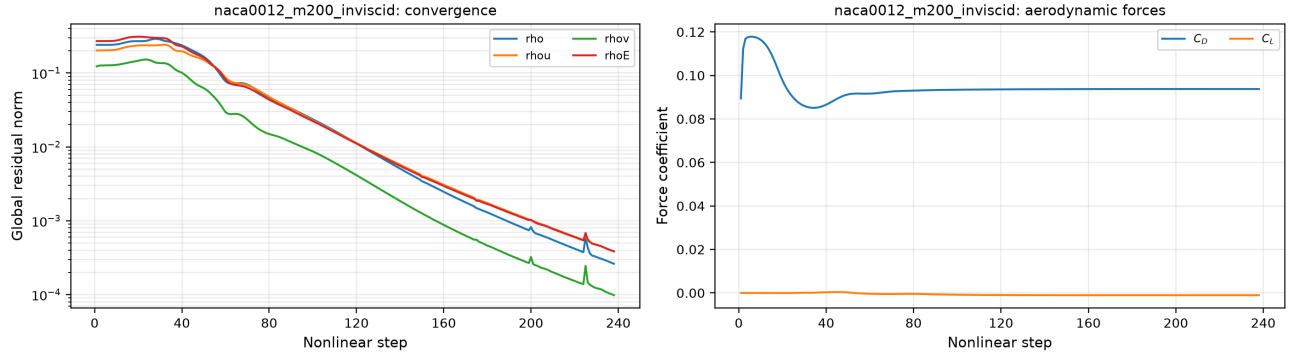


(a) Near-body pressure field; color limits use the 1st–99th percentiles.

Figure 6: naca0012_m080_inviscid computed-output diagnostics

9.5 naca0012_m200_inviscid

This inviscid slip-wall result is recorded as **converged**; the evidence audit is **converged**. The tail/final values are $C_D = 0.09377743$, $C_L = -0.001038561$, and $C_m = -0.00047275$ at $M_\infty = 2$ and $\alpha = 0^\circ$. The residual reduction is 3.00953 orders; the near-zero lift and wall-condition gates provide the symmetry and boundary checks. The production method is `least_squares_piecewise_linear` with `Barth_Jespersen_positivity_fallback`, `Rusanov_local_Lax_Friedrichs` flux, and `implicit_pseudo_time/local_block_Jacobi_with_CFL` integration. The panels fig. 7a, fig. 7b show the submitted source histories or computed fields named in their captions. The panels fig. 8a, fig. 8b show the submitted source histories or computed fields named in their captions.



(a) Global conservative residual history.

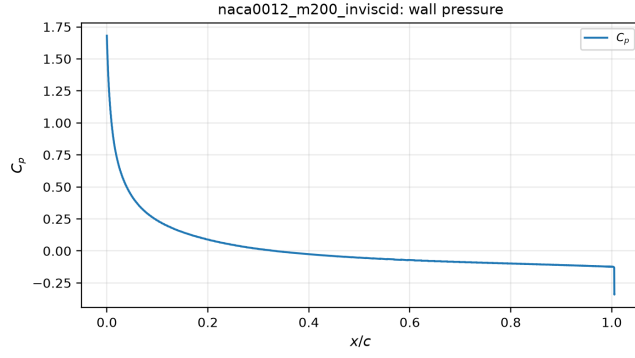
(b) Lift and drag coefficient history.

Figure 7: naca0012_m200_inviscid computed-output diagnostics

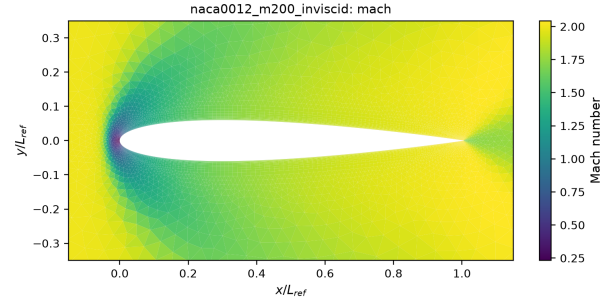
The panels fig. 9a show the submitted source histories or computed fields named in their captions.

9.6 naca0012_m015_laminar_re5000

This laminar no-slip result is recorded as **converged**; the evidence audit is **converged**. The tail/final values are $C_D = 3.18719$, $C_L = -0.001213959$, and $C_m = -0.0007241622$ at $M_\infty = 0.15$ and $\alpha = 0^\circ$. The residual reduction is 4.00553 orders; the near-zero lift and wall-condition gates provide the symmetry and boundary checks. The production method is `least_squares_piecewise_linear` with `Barth_Jespersen_positivity_fallback`, `Rusanov_local_Lax_Friedrichs` flux, and `implicit_pseudo_time/local_block_Jacobi_with_CFL` integration. The panels fig. 10a, fig. 10b show the submitted source histories or computed fields named in their captions. The panels fig. 11a, fig. 11b show the submitted source histories or computed fields named in their captions. The panels fig. 12a, fig. 12b show the submitted source histories or computed fields named in their captions.

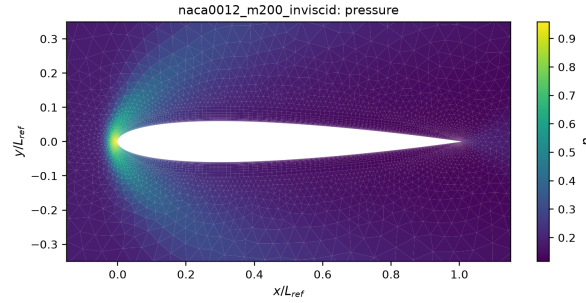


(a) Wall pressure coefficient distribution.



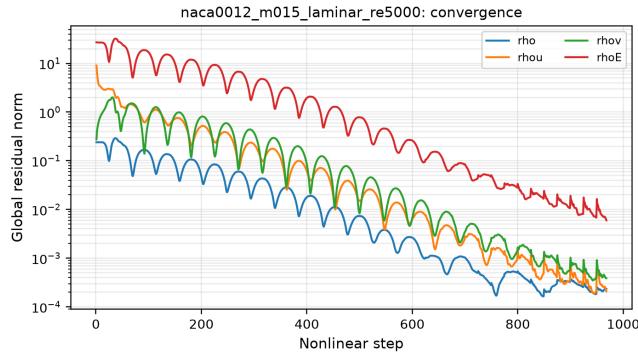
(b) Near-body mach field; color limits use the 1st–99th percentiles.

Figure 8: naca0012_m200_inviscid computed-output diagnostics

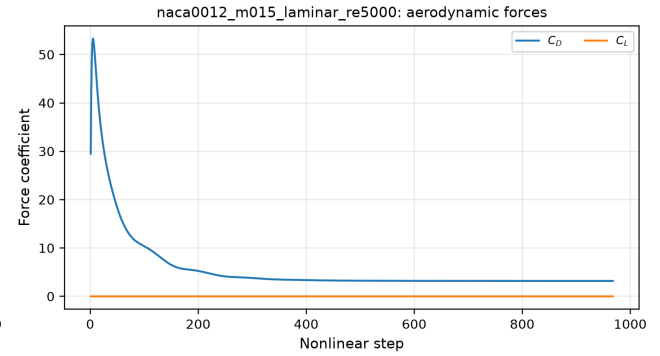


(a) Near-body pressure field; color limits use the 1st–99th percentiles.

Figure 9: naca0012_m200_inviscid computed-output diagnostics

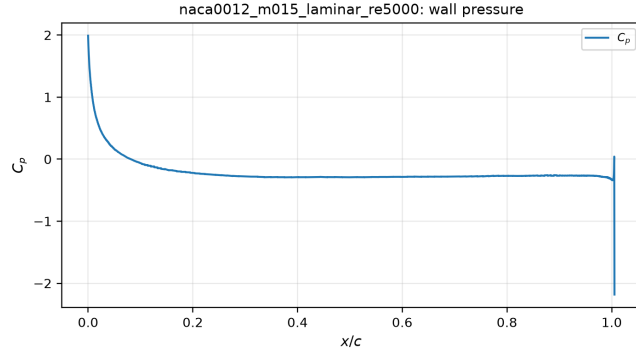


(a) Global conservative residual history.

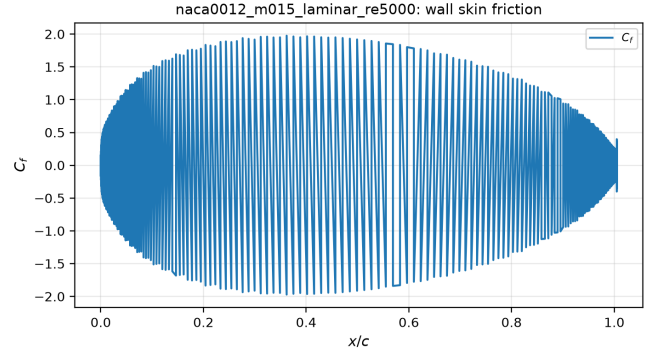


(b) Lift and drag coefficient history.

Figure 10: naca0012_m015_laminar_re5000 computed-output diagnostics

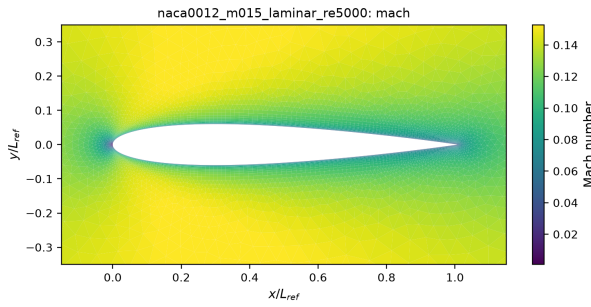


(a) Wall pressure coefficient distribution.

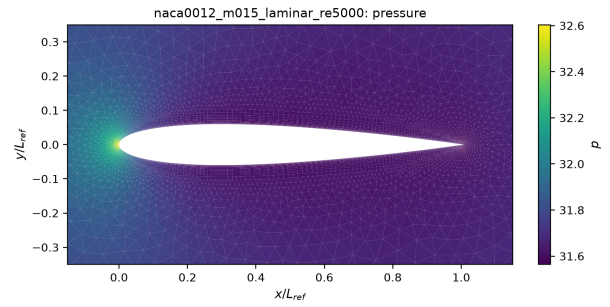


(b) Wall skin-friction coefficient distribution.

Figure 11: naca0012_m015_laminar_re5000 computed-output diagnostics



(a) Near-body mach field; color limits use the 1st–99th percentiles.



(b) Near-body pressure field; color limits use the 1st–99th percentiles.

Figure 12: naca0012_m015_laminar_re5000 computed-output diagnostics

9.7 naca0012_m080_laminar_re5000

This laminar no-slip result is recorded as **converged**; the evidence audit is **converged**. The tail/final values are $C_D = 0.6924408$, $C_L = -0.002168069$, and $C_m = -0.0009840017$ at $M_\infty = 0.8$ and $\alpha = 0^\circ$. The residual reduction is 4.00026 orders; the near-zero lift and wall-condition gates provide the symmetry and boundary checks. The production method is `least_squares_piecewise_linear` with `Barth_Jespersen_positivity_fallback`, `Rusanov_local_Lax_Friedrichs` flux, and `implicit_pseudo_time/local_block_Jacobi_with_CFL` integration. The panels fig. 13a, fig. 13b show the submitted source histories or computed fields named in their captions. The panels fig. 14a, fig. 14b show the submitted source histories or computed fields named in

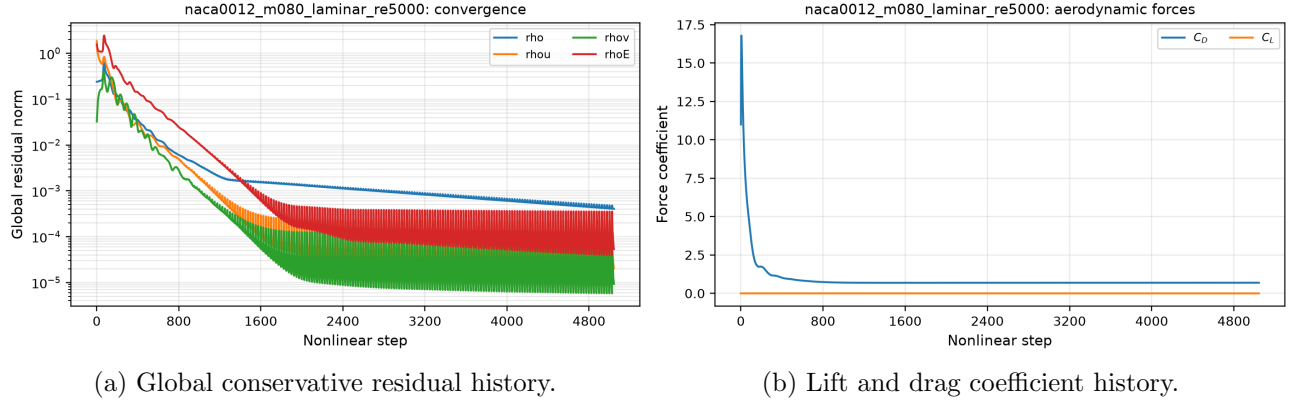


Figure 13: naca0012_m080_laminar_re5000 computed-output diagnostics

their captions. The panels fig. 15a, fig. 15b show the submitted source histories or computed fields named

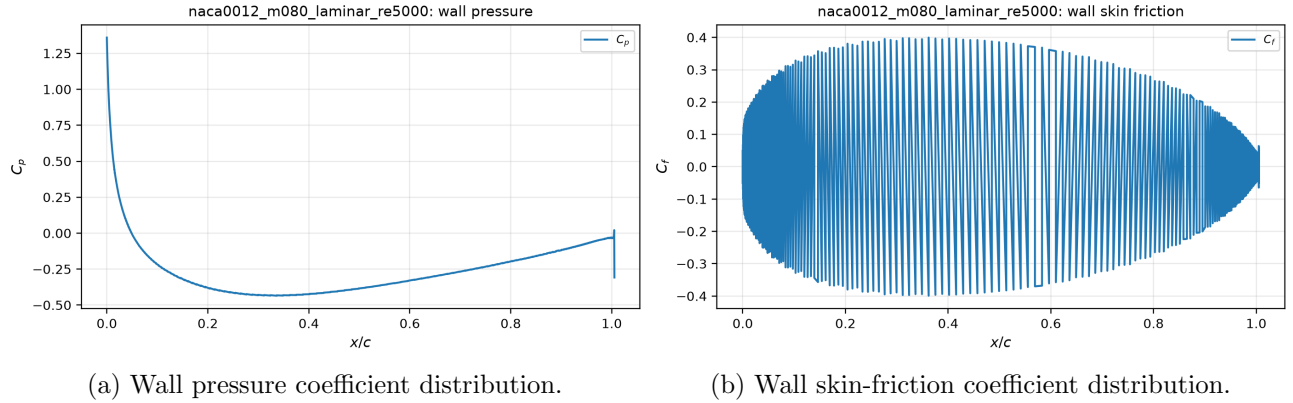
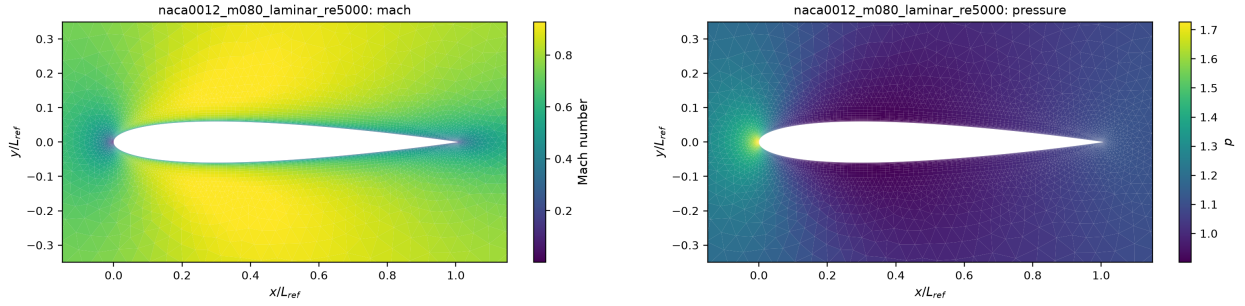


Figure 14: naca0012_m080_laminar_re5000 computed-output diagnostics

in their captions.

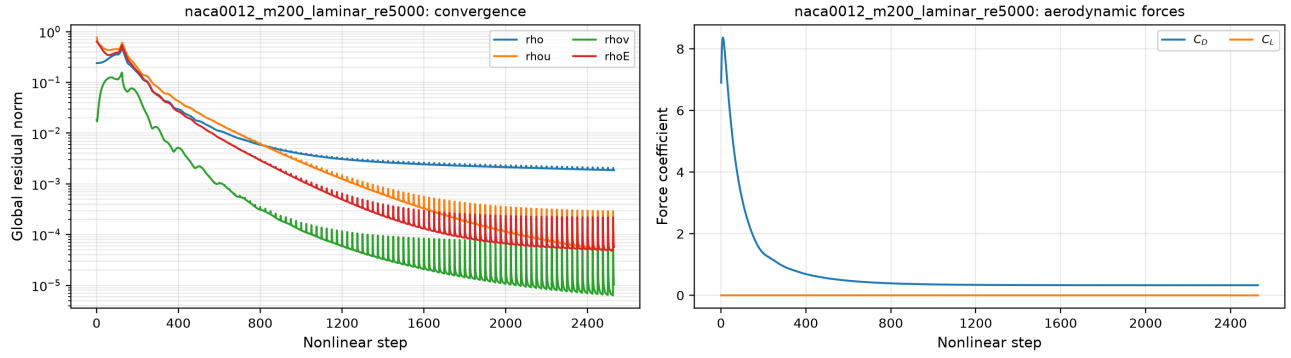
9.8 naca0012_m200_laminar_re5000

This laminar no-slip result is recorded as **converged**; the evidence audit is **converged**. The tail/final values are $C_D = 0.3286775$, $C_L = -0.001271425$, and $C_m = -0.0005967451$ at $M_\infty = 2$ and $\alpha = 0^\circ$. The residual reduction is 3.00002 orders; the near-zero lift and wall-condition gates provide the symmetry and boundary checks. The production method is `least_squares_piecewise_linear` with `Barth_Jespersen_positivity_fallback`, `Rusanov_local_Lax_Friedrichs` flux, and `implicit_pseudo_time/local_block_Jacobi_with_CFL` integration. The panels fig. 16a, fig. 16b show the submitted source histories or computed fields named in their captions. The panels fig. 17a, fig. 17b show the submitted source histories or computed fields named in their captions. The panels fig. 18a, fig. 18b show the submitted source histories or computed fields named



(a) Near-body mach field; color limits use the 1st–99th percentiles. (b) Near-body pressure field; color limits use the 1st–99th percentiles.

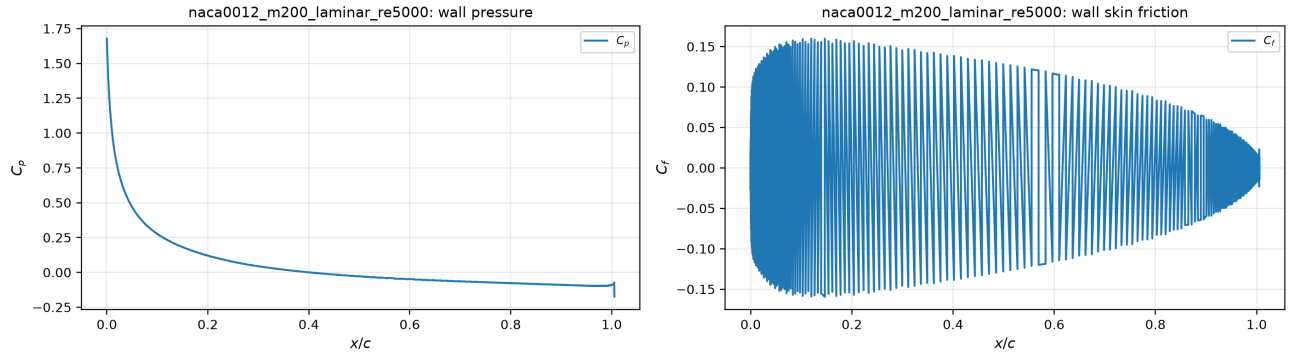
Figure 15: naca0012_m080_laminar_re5000 computed-output diagnostics



(a) Global conservative residual history.

(b) Lift and drag coefficient history.

Figure 16: naca0012_m200_laminar_re5000 computed-output diagnostics

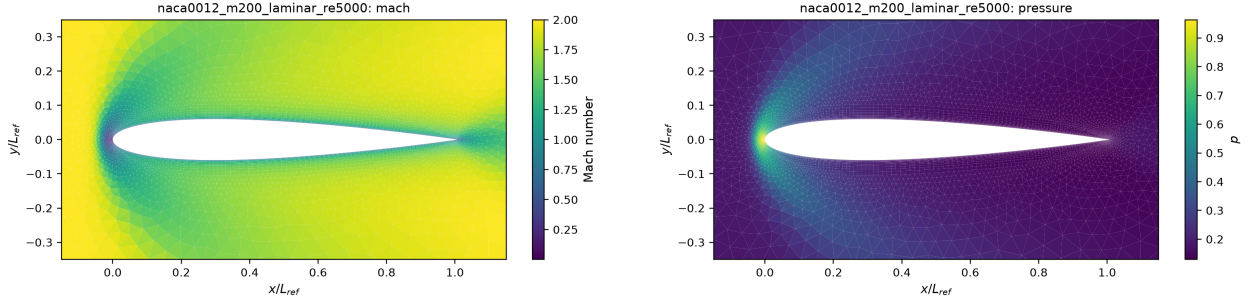


(a) Wall pressure coefficient distribution.

(b) Wall skin-friction coefficient distribution.

Figure 17: naca0012_m200_laminar_re5000 computed-output diagnostics

in their captions.



(a) Near-body mach field; color limits use the 1st–99th percentiles. (b) Near-body pressure field; color limits use the 1st–99th percentiles.

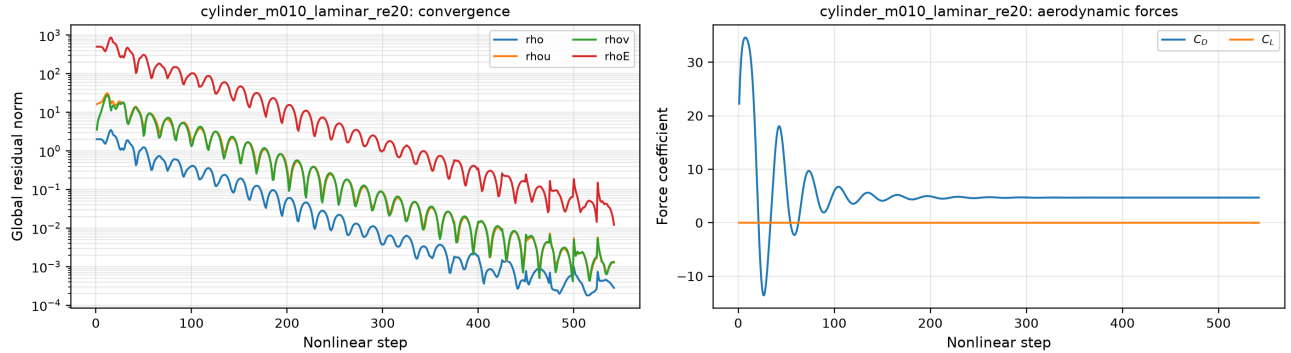
Figure 18: naca0012_m200_laminar_re5000 computed-output diagnostics

9.9 Cylinder Reynolds 20

The Reynolds-20 cylinder is a steady laminar case. Positive tail drag, varying wall pressure, nonzero skin friction, and a bounded residual history support a steady recirculating wake interpretation.

9.10 cylinder_m010_laminar_re20

The result is **converged** with tail $C_D = 4.701114$ and $C_L = -0.001426419$; no-slip wall and positivity checks pass. The wake panel is a computed unstructured-cell rendering. The panels fig. 19a, fig. 19b show the submitted source histories or computed fields named in their captions. The panels fig. 20a,



(a) Global conservative residual history.

(b) Lift and drag coefficient history.

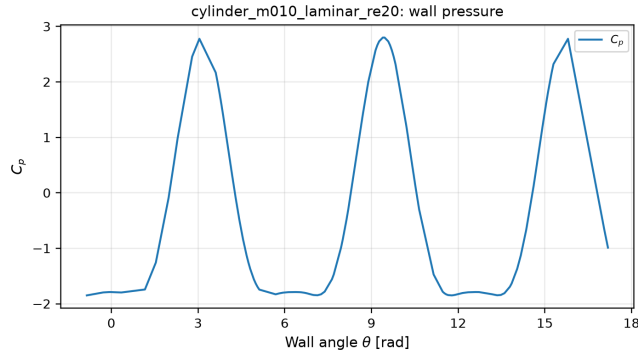
Figure 19: cylinder_m010_laminar_re20 computed-output diagnostics

fig. 20b show the submitted source histories or computed fields named in their captions. The panels fig. 21a, fig. 21b show the submitted source histories or computed fields named in their captions. The panels fig. 22a show the submitted source histories or computed fields named in their captions.

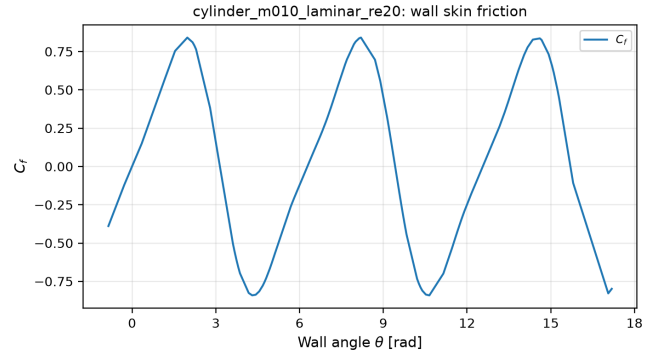
9.11 Cylinder Reynolds 200 Vortex Street

9.12 cylinder_m010_laminar_re200

The $Re = 200$ computation reaches $t = 300$ after 30000 physical steps and is marked **statistically periodic**. The post-transient mean drag is $C_D = 1.274232$, the lift half-range is 0.1741315, and the dominant lift frequency is 0.1444444 in U_∞/L_{ref} units; with the supplied $U_\infty = L_{ref} = 1$, this is $St = 0.1444444$.

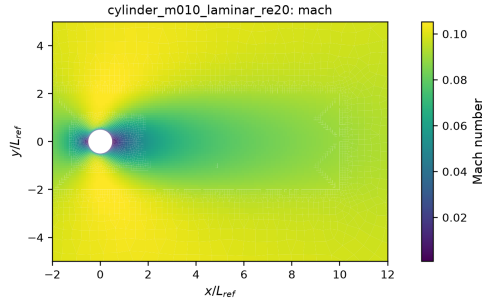


(a) Wall pressure coefficient distribution.

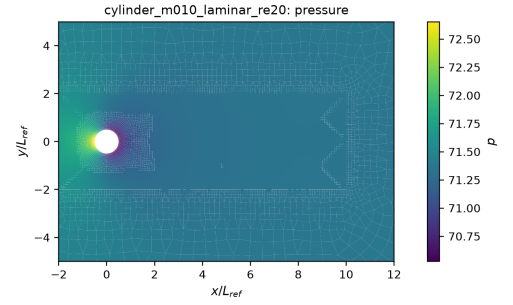


(b) Wall skin-friction coefficient distribution.

Figure 20: cylinder_m010_laminar_re20 computed-output diagnostics

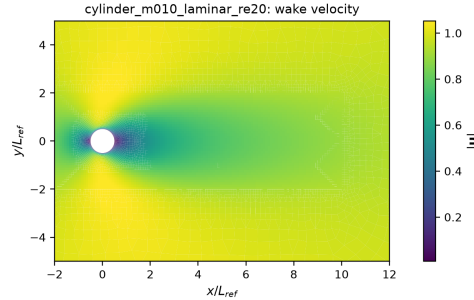


(a) Near-body mach field; color limits use the 1st–99th percentiles.



(b) Near-body pressure field; color limits use the 1st–99th percentiles.

Figure 21: cylinder_m010_laminar_re20 computed-output diagnostics



(a) Post-transient cylinder wake velocity-magnitude field from the computed unstructured cells.

Figure 22: cylinder_m010_laminar_re20 computed-output diagnostics

Observed inner iterations span 8–193 (mean 14.973), with 0 target misses and converged fraction 1. The panels fig. 23a, fig. 23b show the submitted source histories or computed fields named in their captions. The panels fig. 24a, fig. 24b show the submitted source histories or computed fields named in their captions.

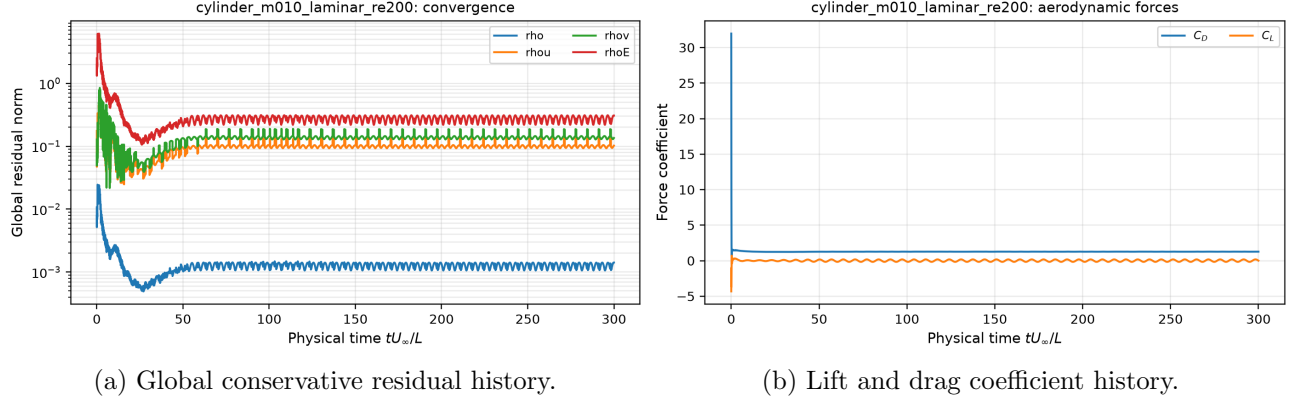


Figure 23: cylinder_m010_laminar_re200 post-transient diagnostic panels

tions. The panels fig. 25a, fig. 25b show the submitted source histories or computed fields named in

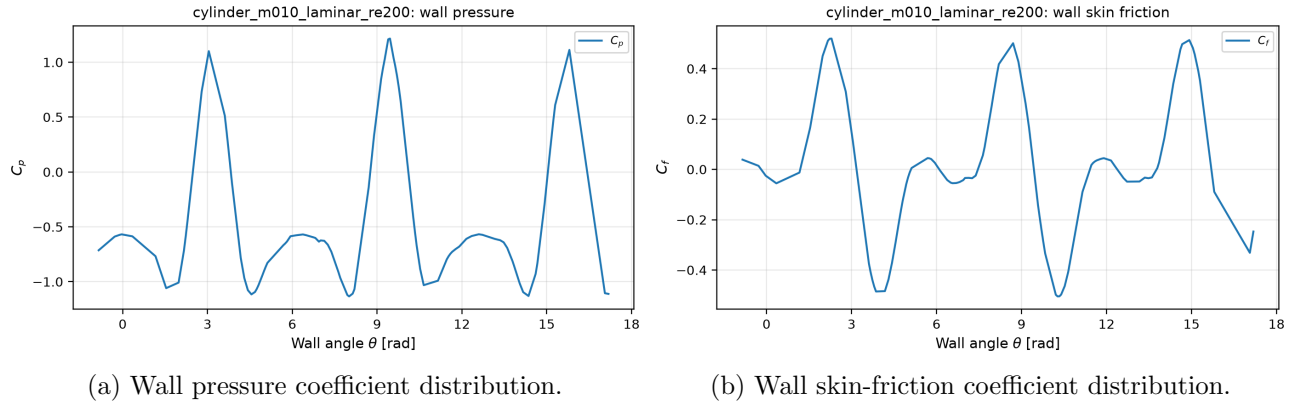


Figure 24: cylinder_m010_laminar_re200 post-transient diagnostic panels

their captions. The panels fig. 26a show the submitted source histories or computed fields named in their captions.

10 Parallel Validation

Table 7 compares independent full-horizon $n_p = 1$ and submitted $n_p = 8$ runs for one NACA and one cylinder case. Final force differences come from the last force rows; timing ratios are machine-specific and should be read with the edge-cut and load-balance values in Table 4.

Table 7: Independent rank-count consistency evidence.

Case	n_p	steps	wall [s]	orders	final C_D	final C_L	status
naca0012_m015_inviscid	1	822	180.184	4.00882	0.0658569	-0.000131017	converged
naca0012_m015_inviscid	8	823	49.3494	4.03211	0.0658558	-0.000130855	converged
cylinder_m010_laminar_re20	1	542	119.431	5.15595	4.70139	-0.00142396	converged

cylinder_m010_laminar_re20	8	542	30.7102	5.15908	4.7014	-0.00142646	converged
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naca0012_m015_inviscid: $|\Delta C_D| = 1.1e - 06$, $|\Delta C_L| = 1.62e - 07$, measured $T_1/T_8 = 3.6512$. cylinder_m010_laminar_re20: $|\Delta C_D| = 1e - 05$, $|\Delta C_L| = 2.5e - 06$, measured $T_1/T_8 = 3.889$.

11 Visualization and Plot Style Requirements

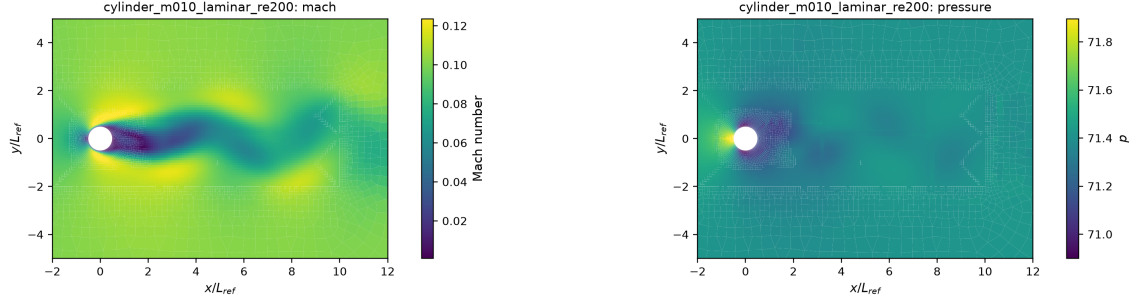
Line plots use labeled nondimensional time or nonlinear-step axes, legends, grid lines, controlled font sizes, and consistent line widths. Field plots use actual unstructured cells or triangulated cells, percentile color limits, variable-specific colorbar labels, near-body NACA windows, and a cylinder wake window. The figure manifest verifies that filenames match variables: **mach** images plot Mach and **pressure** images plot pressure. The Re 200 wake image is the post-transient computed velocity magnitude for this field package; a supplied vorticity array would be clipped to $[-5, 5]$ and identified in its caption.

12 Limitations and Failure Analysis

The solver uses a diffusive Rusanov flux and scalar local block-Jacobi relaxation rather than coupled LU-SGS or Krylov linearization. To maintain robustness on the supplied meshes, steady high-order reconstruction is refreshed every 25th outer step and the transient limiter coefficient is unity; this is a documented accuracy/robustness tradeoff. The final field is gathered on rank zero only after convergence, so output serialization is not a distributed-visualization claim. METIS edge cuts and load ratios are reported explicitly, and all positivity, wall, force, periodicity, MPI, and figure-variable gates remain visible in **sanity_checks.json**. No failed or short diagnostic run is included as a completed production case.

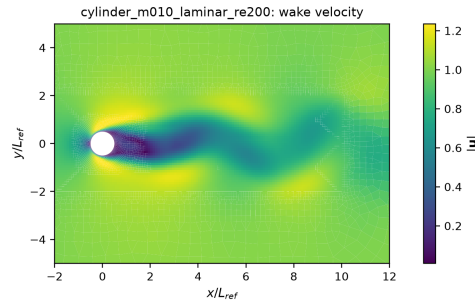
13 Conclusion

The submission contains buildable C++17/MPI source, reproducible case commands, complete output packages for all eight required cases, generated visualizations, machine-readable audits, and this academic LaTeX report. The claims above can be regenerated with **solver/tools/report.py** from the submitted CSV and field files.



(a) Near-body mach field; color limits use the 1st–99th percentiles. (b) Near-body pressure field; color limits use the 1st–99th percentiles.

Figure 25: cylinder_m010_laminar_re200 post-transient diagnostic panels



(a) Post-transient cylinder wake velocity-magnitude field from the computed unstructured cells.

Figure 26: cylinder_m010_laminar_re200 post-transient diagnostic panels